

Lightweight AI for Screen Time Classification

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Motivation

Existing screen time tools measure usage duration of different apps but provide little insight into how users spend time on their devices. They cannot distinguish between work and entertainment activity, limiting their usefulness for self-monitoring and digital wellness [1].

Problem Statement

How can we use a lightweight, privacy-preserving, on-device AI system to analyze screen content and accurately classify user activity as work or entertainment?

Methodology

We developed 2 lightweight models for screen-time classification using normalized 224 x 224 screenshots as input. We also developed an iPadOS application, integrating screenshot capture, on-device inference, and visual feedback.

Model A Neutral v. Non-Neutral

- ❖ MobileNetV3 architecture
- ❖ Binary classification of neutral UI states vs. active content
- ❖ Trained on our iPadOS screenshot dataset
- ❖ Weighted random sampling to address class imbalance between neutral and non-neutral frames

Model B Work v. Entertainment

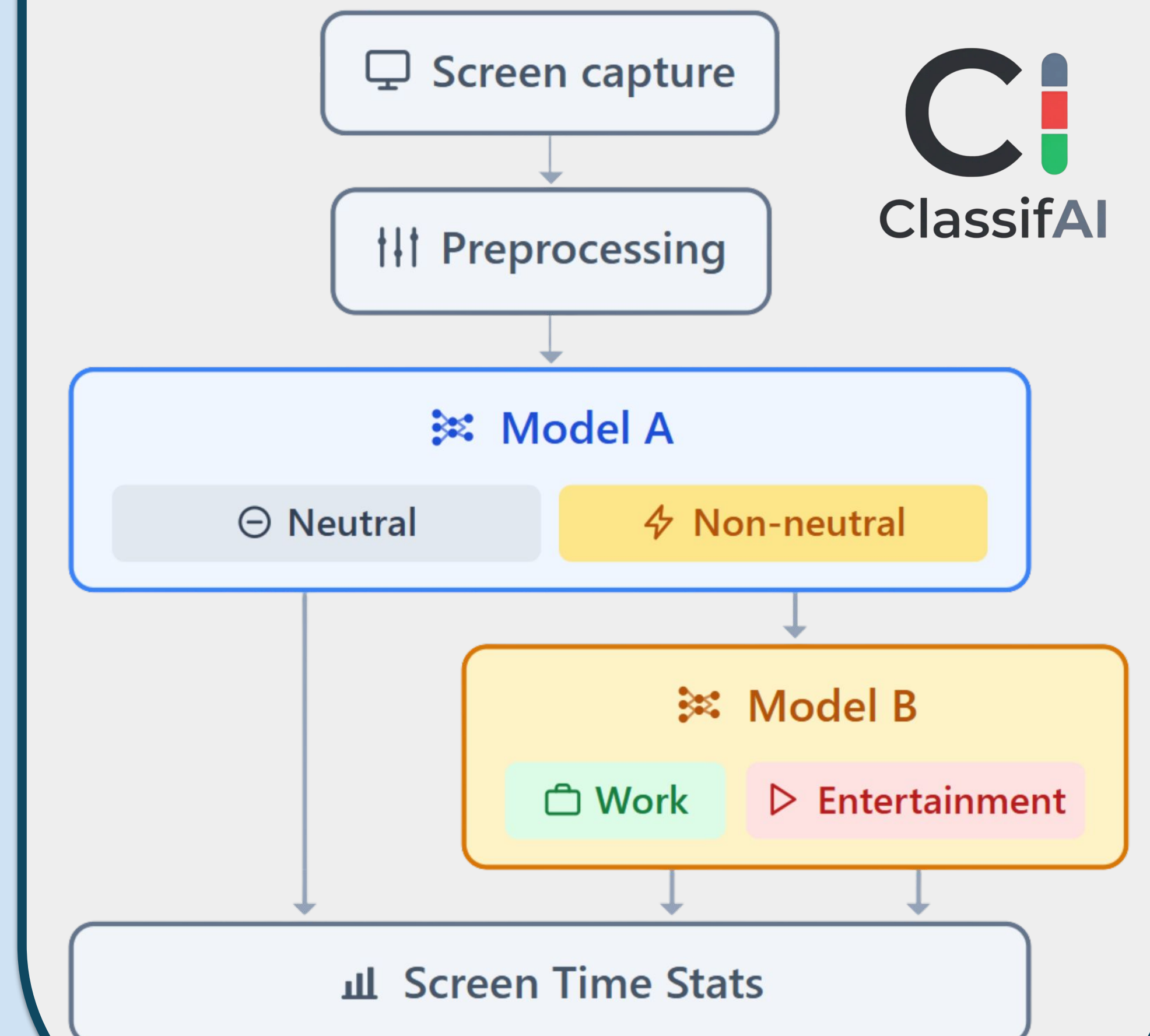
- ❖ MobileNetV3 architecture
- ❖ Binary classification of work vs. entertainment activity from screen content
- ❖ Trained on our iPadOS screenshot dataset
- ❖ Only classifies frames classified as active by Model A as a secondary classification model

ClassifAI iOS Application

- ❖ Built using Flutter and Dart
- ❖ On-device screen capture via ReplayKit
- ❖ CoreML integration for on-device inference
- ❖ Background classification queue to handle session data
- ❖ Screenshots deleted immediately after inference

Our App

Developed a privacy-preserving activity classification app using a two-stage hierarchical vision model pipeline



Results

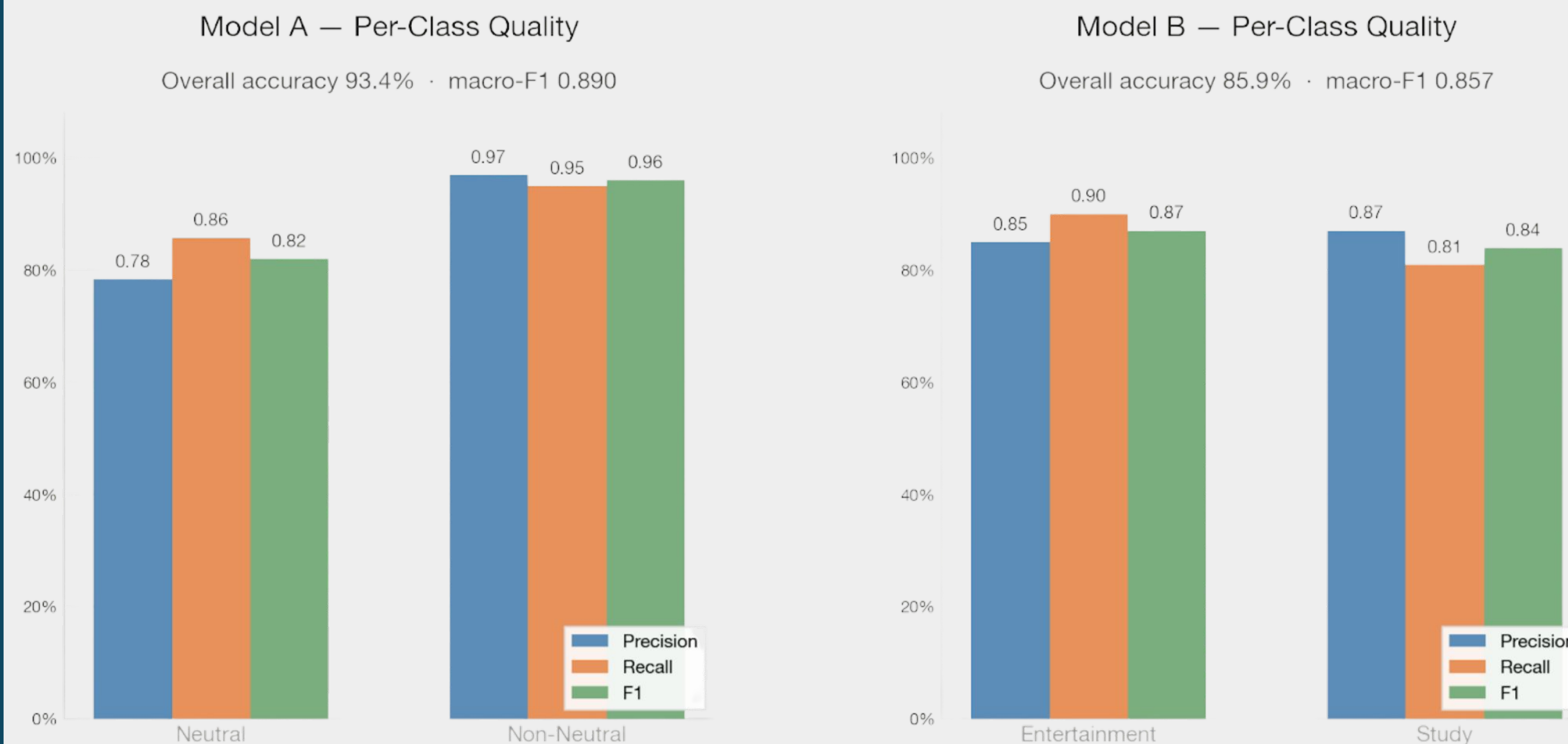


Figure 1. Per-class precision, recall, and F1 scores for Model A (neutral vs. non-neutral classification) and Model B (entertainment vs. work classification), evaluated on held-out benchmark sets with no source overlap between training and test data.

Findings

Our two-stage pipeline achieves classification accuracies of:

93.4% on Model A
85.9% on Model B

Model A misclassified some clean web-app interfaces as neutral due to visual similarity with iOS system screens

Model B shows balanced performance across both classes, with entertainment recall outperforming study recall

Future Work

Model Improvements:

Larger and more diverse training datasets, additional model inputs, different model orderings and combinations, and further optimization of the classification pipeline

App Improvements:

Iterative testing on current features, expanded platform support, additional application features, and cross-platform implementation for more holistic tracking of user screen time

References

- [1] Katevas et al., Typical Phone Use Habits: Intense Use Does Not Predict Negative Well-Being, MobileHCI 2018.

Acknowledgements

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